Project Initialization and Planning Phase

Date	June 28, 2026
Team ID	
Project Title	Credit Card Approval Prediction using Machine Learning
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

This project focuses on automating credit card approval decisions through machine learning. Applicant financial and demographic data is analyzed to predict whether an application should be approved or declined. The system cuts down manual work, brings more consistency, accelerates decisions, and lowers the chance of human error.

Project Overview	
Objective	Design an ML-driven system capable of accurately predicting whether a submitted credit card application merits approval or rejection.
Scope	Covers gathering data, cleaning and preparing it, engineering features, training and evaluating models, and deploying the solution via Flask. Several algorithms are benchmarked to select the strongest performer for live predictions.
Problem Statement	
Description	Banks handle an enormous number of credit card applications on a daily basis. Reviewing them manually takes time, produces inconsistent outcomes, and is susceptible to human error — leading to slower approvals and higher costs.
Impact	An automated approval workflow speeds up decisions, boosts prediction accuracy, lowers financial exposure, cuts manual workload, and leads to happier customers.
Proposed Solution	
Approach	Historical applicant records are cleaned and used to train several ML models — Logistic Regression, Decision Tree, Random Forest, and XGBoost. The top-performing model is then embedded into a Flask web application for live predictions.
Key Features	• Automatic Credit Card Approval Predictions • High Prediction Accuracy via Machine Learning • Live, Real-Time Predictions on the Web • Intuitive, User-Friendly Interface • Rapid Decision Turnaround • Support for Deployment via IBM Watson Machine Learning • Scalable and Secure System Design

Resource Requirements

Resource Type	Description	Specification/ Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i7 Processor (6+ Cores)
Memory	RAM specifications	16 GB RAM (Minimum), 32 GB Recommended
Storage	Disk space for datasets, trained models, and application files	1 TB SSD / 2 TB HDD
Software		
Frameworks	Python Web Framework	Flask
Libraries	Machine Learning and Data Processing Libraries	Scikit-learn, Pandas, NumPy, Seaborn, XGBoost, LightGBM
Development Environment	IDE	Jupyter Notebook, PyCharm, VS Code
Data		
Data	Source, Size, Format	Credit Card Approval Dataset (Kaggle), CSV Format, ~35,000+ Applicant Records